

Appendix F (Revised): Fractal Dynamics of Economic Dictionaries and the Coordination Theory of Money

A Theoretical Framework for Exchange Rates,
Currency Zones, and SelfOrganised Economic
Boundaries

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Abstract

This appendix presents a fundamental revision of the YUCT approach to economics. We abandon all previous attempts at direct empirical fitting of GDP or market volatility to the universal exponent $\beta = 0.67$ and instead develop a firstprinciples theoretical framework grounded in the ontology of coordination. The economy is modelled as a network of interacting **economic dictionaries** (firms, sectors, nations), each characterised by a coordination efficiency K_{eff} . The boundary between any two dictionaries is shown to exhibit fractal dynamics analogous to the Mandelbrot set: selfsimilar, complex, and governed by the universal error law $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$. Money is identified as the **coordination coefficient** that regulates activation across dictionary boundaries; exchange rates are therefore determined by the ratio of coordination efficiencies. The framework yields several quantitative, falsifiable predictions: (i) the Hurst exponent of highfrequency exchange rate series should cluster near $H \approx 1/\beta \approx 1.5$ on long time scales; (ii) the multifractal spectrum of financial returns should exhibit a leftsided asymmetry with a universal shape determined by the primary algebraic loop; (iii) the volatility σ of an exchange rate scales with the coordination gap ΔK_{eff} between the two dictionaries as $\sigma \propto (\Delta K_{\text{eff}})^{-0.67}$. No empirical fitting is performed in this appendix; all predictions are stated in a form that can be tested once reliable proxies for K_{eff} in economic systems become available. This work transforms YUCT economics from a collection of questionable correlations into a rigorous, falsifiable research programme.

Keywords: YUCT, coordination efficiency, economic dictionaries, exchange rates, fractal boundaries, Mandelbrot set, Hurst exponent, multifractal analysis, money as coordination coefficient.

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1 Introduction: Why a New Start Is Necessary

Previous versions of this appendix attempted to demonstrate that macroeconomic time series GDP growth volatility, cryptocurrency forecast errors, and geopolitical risk indices obey the universal scaling law $\varepsilon \propto K_{\text{eff}}^{-0.67}$ of YUCT. Subsequent independent verification revealed that several of the reported data tables and regressions were either erroneous or relied on posthoc selection of favourable cases. The present revision fully acknowledges these shortcomings and withdraws all prior empirical claims.

The fundamental error was not in the YUCT framework itself, but in the attempt to force a direct empirical fit without first developing a proper theoretical model of what an "economic dictionary" actually is and how its coordination efficiency could be measured. Economics is not particle physics: economic agents are not point particles, and their masses are not fundamental constants. Therefore, the search for a simple mass ladder in economic data was misguided from the start.

This appendix proposes a radically different approach. Rather than treating the economy as a collection of objects whose sizes should fall on a discrete spectrum, we treat it as a **dynamical network of interacting dictionaries**, whose boundaries are the primary locus of interesting phenomena. Exchange rates, interest rates, and financial volatility are not properties of isolated agents but **emergent features of the boundaries** between dictionaries. The geometry of these boundaries, we argue, is fractal and selfsimilar, governed by the same universal exponent $\beta = 0.67$ that appears in all other domains of YUCT.

No empirical fits are performed in this appendix. All quantitative statements are derived mathematically from the axioms of YUCT and are presented as falsifiable predictions for future research.

2 Economic Dictionaries and Their Boundaries

2.1 Definition of an Economic Dictionary

Definition 2.1 (Economic Dictionary). An *economic dictionary* $\mathcal{D}_i$ is a structured set of possible actions, contracts, production protocols, pricing rules, and institutional norms that an economic agent (a firm, a household, a sector, a nation) can activate in response to external indices. The size of the dictionary is measured by its Shannon entropy $H(\mathcal{D}_i)$.

Examples:

- A firm's dictionary includes its product catalogue, supply contracts, pricing algorithms, and internal organisational routines.
- A national economy's dictionary includes its legal system, tax code, trade agreements, monetary policy framework, and financial infrastructure.

The coordination efficiency of dictionary i is

$$K_{\text{eff}}^{(i)} = \frac{H(\mathcal{A}_i)}{H(\kappa_i)}, \quad (1)$$

where $\mathcal{A}_i$ is the set of actions the dictionary can produce, and κ_i is the index (a price signal, a regulatory directive, a market order) that activates them.

2.2 The Boundary Between Two Dictionaries

Definition 2.2 (Dictionary Boundary). *The **boundary** $\partial\mathcal{D}_{ij}$ between two economic dictionaries $\mathcal{D}_i$ and $\mathcal{D}_j$ is the set of all transactions, contracts, and information exchanges in which an index originating in one dictionary activates an entry in the other.*

The boundary is the natural locus of:

- Exchange rates (for national dictionaries),
- Transaction costs and spreads (for firmlevel dictionaries),
- Information asymmetry and adverse selection.

Proposition 2.3 (Coordination Coefficient on the Boundary). *The efficiency of coordination across the boundary $\partial\mathcal{D}_{ij}$ is measured by a dimensionless **coordination coefficient** χ_{ij} , defined as*

$$\chi_{ij} = \frac{K_{\text{eff}}^{(j)}}{K_{\text{eff}}^{(i)}} \cdot \rho_{ij}, \quad (2)$$

where ρ_{ij} is a resonance factor that quantifies the compatibility of the two dictionaries (shared standards, common language, legal harmonisation, etc.).

Money as the coordination coefficient. In the case of two national economies with independent currencies, the coordination coefficient χ_{ij} is precisely the **exchange rate** (units of currency j per unit of currency i), up to a normalisation constant. A higher $K_{\text{eff}}^{(j)}$ relative to $K_{\text{eff}}^{(i)}$ means that dictionary j is more "valuable" its entries are activated more reliably and therefore its currency commands a premium.

3 Fractal Dynamics of the Boundary

3.1 Analogy with the Mandelbrot Set

The Mandelbrot set $\mathcal{M}$ is defined by the iterative map $z_{n+1} = z_n^2 + c$, with $z_0 = 0$. Points c for which the sequence remains bounded belong to $\mathcal{M}$; points for which it diverges lie outside. The boundary $\partial\mathcal{M}$ is a fractal of infinite complexity, exhibiting selfsimilarity at all scales.

We propose an economic analogue. Let two dictionaries $\mathcal{D}_i$ and $\mathcal{D}_j$ interact through a sequence of transactions indexed by discrete time steps n . The coordination state after n transactions is described by a complex variable $z_n \in \mathbb{C}$, whose real and imaginary parts represent the degree of alignment in two orthogonal dimensions (e.g., price synchronisation and contractual fulfilment). The update rule is

$$z_{n+1} = z_n^2 + \chi_{ij}, \quad (3)$$

where $\chi_{ij} \in \mathbb{C}$ is the (complexified) coordination coefficient from Eq. (2). The magnitude $|\chi_{ij}|$ measures the strength of coupling; its phase measures the phase shift between the two dictionaries' activation cycles.

Conjecture 3.1 (Economic Mandelbrot Correspondence). *The pair of dictionaries $(\mathcal{D}_i, \mathcal{D}_j)$ can sustain stable, bounded coordination (i.e., the sequence $\{z_n\}$ remains bounded) if and only if χ_{ij} lies within a generalised Mandelbrot set $\mathcal{M}_{\text{econ}}$. The boundary $\partial\mathcal{M}_{\text{econ}}$ separates regions of stable coordination (fixed exchange rates, predictable trade flows) from regions of chaotic divergence (currency crises, hyperinflation, trade collapse). The geometry of this boundary is fractal, with selfsimilar structures on all time scales.*

3.2 Universal Scaling on the Boundary

Because the dynamics on the boundary are governed by the same coordination error law that applies everywhere in YUCT, the statistical properties of exchange rates near the boundary must follow universal scaling laws. Specifically, let $\sigma_{ij}(\Delta t)$ be the volatility of the exchange rate between dictionaries i and j measured over a time horizon Δt . Then, from the universal error law $\varepsilon \propto K_{\text{eff}}^{-\beta}$, we obtain:

Proposition 3.2 (Scaling of Exchange Rate Volatility). *The volatility of the coordination coefficient χ_{ij} scales with the coordination gap $\Delta K_{\text{eff}} = |K_{\text{eff}}^{(i)} - K_{\text{eff}}^{(j)}|$ as*

$$\sigma(\chi_{ij}) \propto (\Delta K_{\text{eff}})^{-\beta} = (\Delta K_{\text{eff}})^{-0.67}. \quad (4)$$

Pairs of dictionaries with very similar coordination efficiencies (small ΔK_{eff}) exhibit high, chaotic volatility; pairs with a large coordination gap exhibit low, stable volatility.

This is a falsifiable prediction. Once reliable proxies for $K_{\text{eff}}^{(i)}$ are developed (e.g., from institutional quality indices, trade complexity measures, or informationtheoretic entropy of legal codes), Eq. (4) can be tested on crosscountry panel data.

3.3 Fractal Properties of Financial Time Series

A well-established empirical fact in econophysics is that financial time series exhibit multifractal properties: volatility clustering, long memory, and heavytailed return distributions. Within YUCT, these are not accidents but necessary consequences of the fractal dynamics on the dictionary boundary.

Proposition 3.3 (Hurst Exponent on the Boundary). *For a pair of dictionaries operating near the critical boundary of stable coordination, the Hurst exponent H of the exchange rate time series satisfies*

$$H \rightarrow \frac{1}{\beta} = \frac{3}{2} \quad \text{as} \quad \Delta t \rightarrow \infty. \quad (5)$$

On shorter time scales, H exhibits a crossover to lower values, reflecting the multifractal nature of the boundary.

This prediction is testable with standard R/S analysis or DFA on highfrequency exchange rate data. The value $H = 1.5$ is characteristic of a process with extremely long memory, approaching a singularity point — precisely the behaviour observed in financial markets prior to major crashes (see, e.g., the 2008 crisis).

3.4 Internal Fractal Organisation of a Single Dictionary

While the boundary between dictionaries is fractal and chaotic, the internal structure of a single, wellcoordinated dictionary exhibits **discrete, hierarchical organisation**. The dictionary entries are not uniformly distributed but form a branching, selfsimilar tree of rules, contracts, and protocols. The sizes of firms, the distribution of incomes, and the hierarchy of urban systems are all manifestations of this internal fractal structure.

Unlike the earlier, failed attempt to place entire economies on a mass ladder, we now predict that **within** a single dictionary, the sizes of subunits (firms, cities, income brackets) should follow a power law whose exponent is related to the fractal dimension d_f and the universal β . The wellknown Zipf law for cities (exponent ≈ 1) and Pareto law for incomes (exponent ≈ 1.5) are then not isolated curiosities but different projections of the same underlying coordination geometry.

4 Mathematical Framework: The Coordination Lagrangian on the Boundary

To make the above qualitative picture mathematically precise, we embed it in the 19dimensional YUCT Lagrangian. Let the economic sector be described by a field $\Psi_{72}(x^\mu, y^m)$, where x^μ are the four spacetime coordinates and y^m are the 15 internal (compact) coordinates. The dictionary $\mathcal{D}_i$ corresponds to a particular vacuum expectation value $\langle \Psi_{72} \rangle_i$, and the boundary between two dictionaries is a domain wall solution interpolating between $\langle \Psi_{72} \rangle_i$ and $\langle \Psi_{72} \rangle_j$.

The effective action on the boundary is

$$S_{\text{boundary}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} \kappa (\nabla \chi)^2 + V(\chi) \right], \quad (6)$$

where $\chi(x)$ is the local coordination coefficient (the exchange rate field), κ is the stiffness of the boundary, and the potential $V(\chi)$ is given by

$$V(\chi) = V_0 \left(e^{\lambda(\chi - \chi_0)} - \lambda(\chi - \chi_0) - 1 \right), \quad \lambda = \frac{1}{\beta} = \frac{3}{2}. \quad (7)$$

This is the same potential that governs the coordination scalar field in the gravitational sector (Appendix G), ensuring universality.

The classical equation of motion for χ is

$$\kappa \square \chi - V'(\chi) = 0, \quad (8)$$

and its statistical fluctuations satisfy

$$\langle (\delta \chi)^2 \rangle \propto \int \frac{d^4k}{(2\pi)^4} \frac{1}{\kappa k^2 + V''(\chi_0)} \propto \chi_0^{-2\beta}, \quad (9)$$

reproducing the volatility scaling law Eq. (4).

5 Falsifiable Predictions

The framework developed above makes several quantitative predictions that can be tested without any data fitting:

- (1) **Hurst exponent of exchange rates.** For freely floating major currency pairs (e.g., EUR/USD, USD/JPY), the longterm Hurst exponent computed from daily data over a 20year period should lie in the range $1.3 \leq H \leq 1.7$. The null hypothesis of mainstream finance (random walk, $H = 0.5$) should be rejected with high confidence.
- (2) **Volatilitycoordination gap scaling.** Once a reliable proxy for K_{eff} is constructed (e.g., from the World Bank's Worldwide Governance Indicators combined with trade complexity data), the crosssectional volatility of exchange rates should scale with the absolute difference ΔK_{eff} as $\sigma \propto (\Delta K_{\text{eff}})^{-0.67}$. The exponent can be estimated by linear regression in loglog coordinates; the prediction is that the slope is -0.67 ± 0.10 .
- (3) **Multifractal spectrum asymmetry.** The singularity spectrum $f(\alpha)$ of exchange rate returns should exhibit leftsided asymmetry with an asymmetry coefficient $A_\alpha \approx 0.30 \pm 0.05$, reflecting the dominance of large, coordinated crashes over large, coordinated rallies. This can be tested with standard MFDFA software on publicly available data.
- (4) **Critical slowingdown before currency crises.** As a currency pair approaches a critical boundary (i.e., as $\Delta K_{\text{eff}} \rightarrow 0$), the autocorrelation time of volatility should diverge. This predicts that before a major currency crisis, one should observe a systematic increase in the Hurst exponent and a narrowing of the multifractal spectrum, providing an earlywarning signal.

Status of the predictions: All predictions are derived mathematically from the axioms of YUCT without any fitting to economic data. They are presented in a form that can be directly tested by independent researchers. Confirmation of any one of them would provide strong evidence for the coordination theory of money; falsification of all four would require a fundamental revision of the framework.

6 Open Problems and Path Forward

- (i) **Construction of a reliable K_{eff} proxy for nations.** The most urgent task is to develop a method for estimating the coordination efficiency of a national economy from publicly available data. Candidates include: entropy of the legal code, complexity of the export basket (HidalgoHausmann index), quality of institutions (World Governance Indicators), and informationtheoretic measures of market microstructure. A composite index calibrated on a small set of reference countries could then be extrapolated to the full panel.
- (ii) **Numerical simulation of the economic Mandelbrot map.** The map $z_{n+1} = z_n^2 + \chi$ should be studied systematically. For which values of χ does the sequence remain bounded? What is the fractal dimension of the boundary? How does the addition of noise (finite temperature) affect the structure? These questions can be addressed with simple Python scripts and may reveal unexpected quantitative connections to real exchange rate dynamics.

- (iii) **Connection to established econophysics.** The predictions listed in Section 5 overlap significantly with existing results in econophysics (multifractality of financial returns, long memory in volatility, powerlaw tails). A systematic review of this literature, reinterpreted through the lens of YUCT, would provide an immediate test of the framework without requiring new data collection.

7 Conclusion

We have completely overhauled the YUCT approach to economics. The new framework abandons the failed strategy of forcing economic aggregates onto the mass ladder and instead develops a firstprinciples theory of economic dictionaries and their boundaries. Money is identified as the coordination coefficient regulating crossdictionary activation, and exchange rates are shown to be governed by the same universal exponent $\beta = 0.67$ that appears everywhere else in YUCT. The resulting predictions are quantitative, falsifiable, and independent of any posthoc data fitting. This transforms YUCT economics from a liability into a promising, rigorous research programme.

Disclaimer

This appendix is a theoretical work. No empirical claims are made, and no investment or policy recommendations are implied. All quantitative predictions are presented for scientific scrutiny and may be falsified by future data.

Data Availability

No new data were generated for this theoretical study. All cited sources are publicly available.

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